Trajectory Forecasting for Autonomous Vehicle on Highway: A Graph Neural Network Based Approach

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Introduction

Accurate trajectory forecasting is essential in autonomous driving to ensure safety and efficient navigation in complex highway scenarios.

Vehicles interact in complex ways — they brake, merge, or switch lanes based on nearby agents.

Predicting where each vehicle, especially the ego vehicle, will be essential to avoid collisions and make safe decisions.

- Traditional physics-based models computationally efficient but fall short in capturing complex road behavior.
- RNNs improved motion modeling, but treat each vehicle as an isolated sequence. Not captures explicit spatial structure.
- Graph-based neural networks (e.g., GAT) enable more flexible modeling of inter-agent relationship.

Problem Statement

Goal: Predict the future trajectory of an ego vehicle using past observations of traffic scenes on a highway.

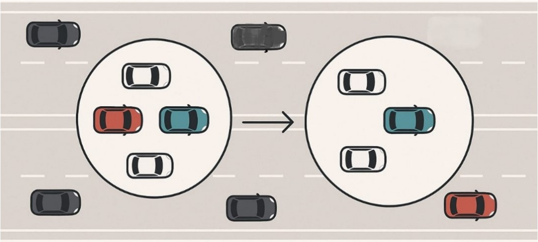
Approach:

Traffic interactions in each timestep of an ego vehicle modeled as a graph:

- **Nodes:** Vehicles are considered as node of graph.
- **Edges:** Proximity-based connection with the ego vehicle.

Each graph captures the spatial structure at a single time step.

Objective: Forecast the future positions (x, y) of the ego vehicle.



Problem Statement

Mathematical description:

Formally, at each time step t, a set of vehicles present in a scene.

Each vehicle i is associated with a feature vector $x_t^i \in \mathbb{R}^F$.

A dynamic interaction graph $G_t = (V_t, E_t)$ is constructed to represent agent relationships based on spatial proximity.

A sequence of graphs $\{G_1, \ldots, G_t\}$, the task is to predict the future positions of the ego vehicle for the next H time steps:

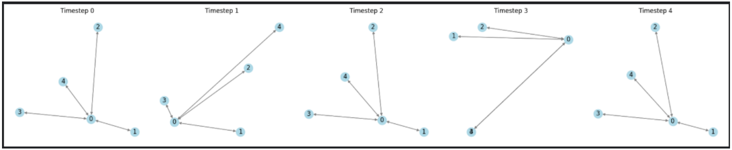
$$\hat{y} = \{\hat{y}_{t+1}, \ldots, \hat{y}_{t+H}\}, \quad \hat{y}_{t+h} \in \mathbb{R}^2$$

where each $\hat{y}_{t+h}$ represents the predicted (x, y) location of the ego vehicle. The model must learn a mapping:

$$f : \{G_1, \ldots, G_t\} \rightarrow \{\hat{y}_{t+1}, \ldots, \hat{y}_{t+H}\}$$

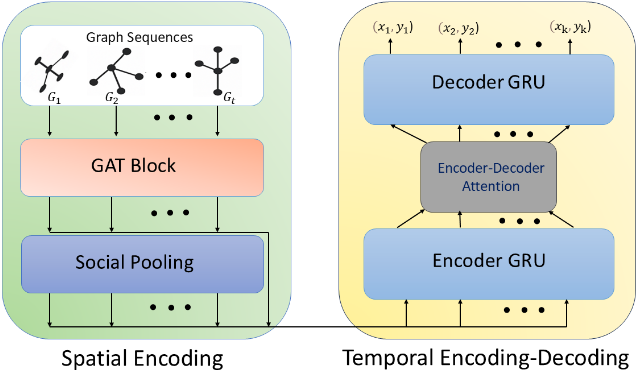
Dataset and Preprocessing

- Dataset: NGSIM US-101 — high-resolution (10 Hz) vehicle trajectory recordings over a 640-meter segment of the US-101 highway in Los Angeles, California.
- Graph construction: nodes = vehicles, edges = proximity.
- Features: Local_X, Local_Y, v_Vel, v_Acc, Lane_ID.
- Normalized to ego-relative frame.
- Output: graph sequence and future coords.



Data processing steps: GitHub Repository

Proposed GAT-GRU model



Spatial Encoding with GAT

At each time step t, a graph $G_t = (V_t, E_t)$ represents the scene. Each node $i \in V_t$ has features $\mathbf{x}_i^t \in \mathbb{R}^F$.
The GAT computes attention-based spatial embeddings as:

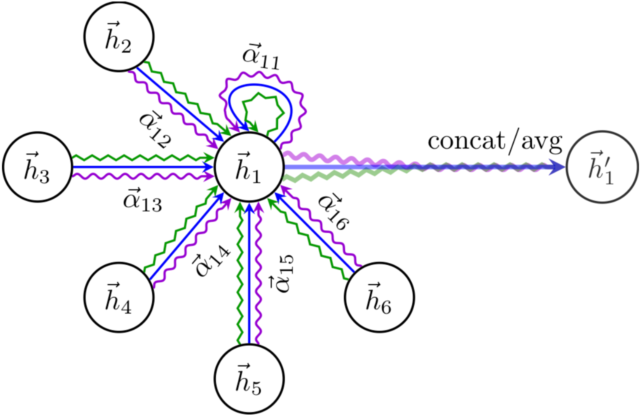
$$e_{ij} = \text{LeakyReLU}\left(\mathbf{a}^\top [\mathbf{W}\mathbf{x}_i \,\|\, \mathbf{W}\mathbf{x}_j]\right), \tag{1}$$

$$\alpha_{ij} = \frac{\exp(e_{ij})}{\sum_{k \in \mathcal{N}(i)} \exp(e_{ik})}, \tag{2}$$

$$\mathbf{h}_i = \sum_{j \in \mathcal{N}(i)} \alpha_{ij} \mathbf{W}\mathbf{x}_j, \tag{3}$$

where $\mathbf{W} \in \mathbb{R}^{d \times F}$ and $\mathbf{a} \in \mathbb{R}^{2d}$ are learnable parameters, and $\mathcal{N}(i)$ denotes the neighbors of node i. The embedding for the ego vehicle $\mathbf{h}_{\text{ego}}$ is extracted for further processing.

Spatial Encoding with GAT



Spatial Encoding with Scocial Pooling

To model social interactions, we apply a learned pooling function $\phi(\cdot)$ over the neighbors of the ego vehicle:

$$\mathbf{z}_{\text{pool}} = \frac{1}{|\mathcal{N}_{\text{ego}}|} \sum_{j \in \mathcal{N}_{\text{ego}}} \phi(\mathbf{h}_j), \tag{4}$$

$$\phi(\cdot) = \text{ReLU}\left(\mathbf{W}_2 \, \text{ReLU}(\mathbf{W}_1 \cdot)\right), \tag{5}$$

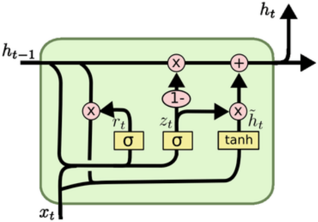
and concatenate this with the ego embedding:

$$\mathbf{z}_t = [\mathbf{h}_{\text{ego}} \,\|\, \mathbf{z}_{\text{pool}}] \in \mathbb{R}^{2d}. \tag{6}$$

Temporal Encoding – GRU

The temporal dynamics of the ego vehicle are encoded using a GRU over the sequence $\{\mathbf{z}_1, \ldots, \mathbf{z}_t\}$:

$$\mathbf{h}_t^{\text{enc}} = \text{GRU}(\mathbf{z}_t, \mathbf{h}_{t-1}^{\text{enc}}), \tag{7}$$



$$z_t = \sigma\left(W_z \cdot [h_{t-1}, x_t]\right)$$
$$r_t = \sigma\left(W_r \cdot [h_{t-1}, x_t]\right)$$
$$\tilde{h}_t = \tanh\left(W \cdot [r_t * h_{t-1}, x_t]\right)$$
$$h_t = (1 - z_t) * h_{t-1} + z_t * \tilde{h}_t$$

where the final hidden state $\mathbf{h}_T^{\text{enc}}$ summarizes the trajectory history.

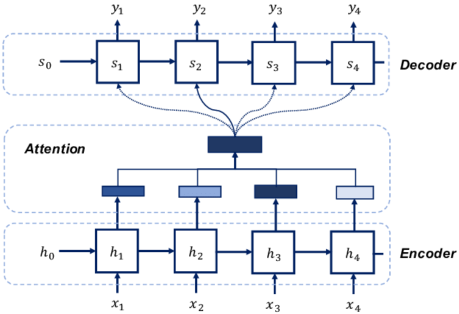
Encoder-Decoder Attention

To allow the decoder to attend to relevant past states, we use an encoder-decoder attention mechanism. At each decoder step h, a context vector $\mathbf{c}_h$ is computed as:

$$e_{h,t} = \frac{\mathbf{q}_h^\top \mathbf{h}_t^{\text{enc}}}{\sqrt{d}} \tag{8}$$

$$\alpha_{h,t} = \frac{\exp(e_{h,t})}{\sum_{t'=1}^{T} \exp(e_{h,t'})} \tag{9}$$

$$\mathbf{c}_h = \sum_{t=1}^{T} \alpha_{h,t} \mathbf{h}_t^{\text{enc}} \tag{10}$$



where $\mathbf{q}_h$ is the decoder query vector, typically the previous decoder hidden state.

Decoder and Output

The decoder GRU is initialized with $\mathbf{h}_T^{\text{enc}}$ and runs autoregressively. At each future step $h \in \{1, \ldots, H\}$:

$$\mathbf{x}_h^{\text{dec}} = [\hat{\mathbf{y}}_{t+h-1} \,\|\, \mathbf{c}_h], \tag{11}$$

$$\mathbf{h}_{t+h}^{\text{dec}} = \text{GRU}(\mathbf{x}_h^{\text{dec}}, \mathbf{h}_{t+h-1}^{\text{dec}}), \tag{12}$$

$$\hat{\mathbf{y}}_{t+h} = \mathbf{W}_o \mathbf{h}_{t+h}^{\text{dec}} + \mathbf{b}_o, \tag{13}$$

where $\hat{\mathbf{y}}_{t+h} \in \mathbb{R}^2$ is the predicted position. During training, ground truth positions are used in place of predictions with a probability p (teacher forcing)

Loss Functions

The model is trained using Mean Squared Error (MSE) and Huber loss:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{H} \sum_{h=1}^{H} \|\hat{\mathbf{y}}_{t+h} - \mathbf{y}_{t+h}\|_2^2, \tag{14}$$

$$\mathcal{L}_{\text{Huber}} = \frac{1}{H} \sum_{h=1}^{H} \text{Huber}(\hat{\mathbf{y}}_{t+h}, \mathbf{y}_{t+h}) \tag{15}$$

During evaluation, we report Average Displacement Error (ADE) and Final Displacement Error (FDE).

$$\text{ADE} = \frac{1}{H} \sum_{h=1}^{H} \|\hat{y}_{t+h} - y_{t+h}\|_2 \tag{16}$$

$$\text{FDE} = \|\hat{y}_{t+H} - y_{t+H}\|_2 \tag{17}$$

These metrics provide intuitive measures of how close the predicted trajectory is to the actual future path.

Experimental Setup

- 80/20 train-validation split, Batch size = 1
- Framework: PyTorch Geometric.
- Hardware: NVIDIA K80 GPU from Kaggle
- Hyperparameters:

Parameter	**Value**
GAT hidden dimension	64
GRU hidden dimension	64
Learning rate	1×10^{-3}
Epochs	20, 30
Optimizer	Adam
Dropout	0.3, 0.4
Weight initialization	Random , Xavier
Weight Decay	1×10^{-5}
Teacher forcing	0.5
Loss Function	MSE, Huber Loss

Results

We compare our proposed **GAT-GRU** model with baseline models:

- *Constant Velocity (CV):* Extrapolates positions assuming constant speed; yields high error.
- *Simple GRU:* Uses only past ego trajectory; moderately better than CV.
- *MLP:* Learns from past coordinates; lacks sequential modeling.
- *GAT-GRU:* Integrates spatial (GAT) and temporal (GRU) info; performs best.

Table: **Performance comparison (Mean Squared Error based metrics)**

Model	**ADE (m)**	**FDE (m)**
Constant Velocity	834.69	1430.25
Simple GRU	328.23	137.27
MLP	312.11	214.75
GAT-GRU	**95.21**	**107.20**

Ablation Study

Table: Ablation study of GAT-GRU model (values in meters).

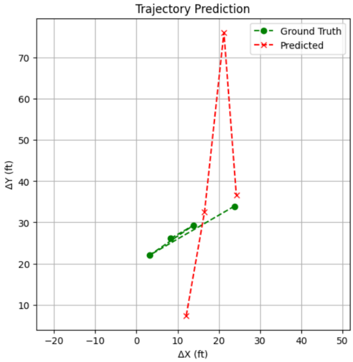
Model Variant	**ADE (m)**	**FDE (m)**
GAT-GRU (Baseline)	**95.22**	**107.20**
+ Social Pooling	103.88	111.16
+ Social Pooling + Attention	112.05	112.22

Observations:

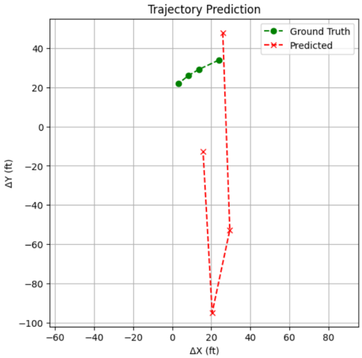
- Social pooling introduces interaction modeling but slightly increases error — possible overfitting due to limited data.
- Temporal attention yields no further improvement — short prediction horizon or undertraining may be factors.
- The core GAT-GRU model performs best, validating its spatio-temporal encoding strength.

Visualization

Predicted vs. ground truth trajectories of an ego vehicle.



Example - 1



Example - 2

Practical Implications

- **Motion Planning Integration**
  Forecasted trajectories guide AV decision-making (lane changes, braking).
- **Safety Considerations**
  Prediction errors can lead to collisions; evaluate with collision-risk metrics.
- **Sim-to-Real Validation**
  Test in simulators like CARLA to verify closed-loop performance.
- **Uncertainty Handling**
  Current model is deterministic; future work includes probabilistic outputs and confidence estimation.

Conclusion & Future Work

Key Takeaways:

- Proposed a GAT-GRU based encoder-decoder model for vehicle trajectory forecasting.
- Integrated social pooling and temporal attention for enhanced spatio-temporal reasoning.
- Outperformed baselines (CV, GRU, MLP) on the NGSIM highway dataset with up to 88% lower ADE.

Limitations:

- Deterministic outputs — does not model uncertainty behaviors.
- Evaluation limited to U.S. highway data; lacks generalization.
- Binary edge definitions — ignore interaction strength.
- The model takes 5 past frames as input and predicts trajectories for the next 4 future frames.

Conclusion & Future Work

Future Work:

- Extend to probabilistic or multimodal forecasting (e.g., CVAE, GAN).
- Incorporate edge features like time-to-collision and relative speed.
- Validate in closed-loop simulation environments (e.g., CARLA,..).
- Train and evaluate on diverse datasets (e.g., Waymo, nuScenes).
- Include extending this to use the past 5 seconds of data to predict 3 seconds into the future.
- Distance threshold for graph construction as hyperparameter for better capture spatial interaction
- Maximum number of vehicles under the distance threshold are allowed for graph construction

References

- Sheng, Z., Xu, Y., Xue, S., & Li, D. (2021). "Graph-Based spatio-temporal Convolutional Network for Vehicle Trajectory Prediction in Autonomous Driving". IEEE Transactions on Intelligent Transportation Systems, 23, 17654-17665.
- K. Zhang, L. Zhao, C. Dong, L. Wu and L. Zheng, "AI-TP: Attention-Based Interaction-Aware Trajectory Prediction for Autonomous Driving", IEEE Transactions on Intelligent Vehicles, vol. 8, no. 1, pp. 73-83, Jan. 2023, doi: 10.1109/TIV.2022.3155236.
- Gao, Y., Fu, J., Feng, W., Xu, T., & Yang, K. " Surrounding vehicle trajectory prediction under mixed traffic flow based on graph attention network." Physica A: Statistical Mechanics and its Applications (2024).
- Ye, C., & Chen, D. "Trajectory Prediction for Autonomous Driving of New Energy Vehicles based on Graph Transformer Networks." Scientific Journal of Intelligent Systems Research. (2025)
- Deo, N.; Trivedi, M.M. "Convolutional Social Pooling for Vehicle Trajectory Prediction". Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR) Workshops, Salt Lake City, UT, USA, 18–23 June 2018.
- A. Alahi, K. Goel, V. Ramanathan, A. Robicquet, L. Fei-Fei, and S. Savarese, "Social LSTM: Human trajectory prediction in crowded spaces", Proc. of the IEEE Conf. Comput. Vis. Pattern Recognition. (CVPR) , 2016, pp. 961–971.

Thank You!